Food Waste Level Prediction System

Applied Machine Learning Project & Technical Evaluation Report

1. Executive Summary & Problem Statement

Commercial food services, university cafeterias, and hotel banquet operations face substantial challenges in optimizing batch cooking volumes. Discrepancies between prepared food and dynamic customer demand generate preventable food waste, resulting in financial loss and elevated carbon footprints. This project implements an end-to-end Machine Learning solution that predicts operational food waste levels into three discrete, actionable categories: **Low Waste (<= 12%)**, **Medium Waste (12% - 25%)**, and **High Waste (> 25%)**.

2. Dataset Description & Operational Target Formulation

Dataset Disclosure: A statistically simulated benchmark dataset (*dataset.csv*) of **3,000 observations** was developed using fixed seed random_seed=42. The simulation is grounded in published commercial catering standards (WRAP Hospitality & FAO Catering Waste Frameworks), capturing diner portion demands across four meal types: Breakfast (0.35-0.48 kg/person), Lunch (0.50-0.68 kg/person), Dinner (0.65-0.85 kg/person), and Buffet/Events (0.75-1.05 kg/person).

Feature Name	Type	Value Range	Operational Definition
meal_type	Categorical	Breakfast, Lunch, Dinner, Buffet	Meal service context
food_prepared_kg	Numerical	5.0 - 500.0 kg	Total food cooked for service
meals_served	Numerical	10 - 450 diners	Actual diner attendance
day_of_week	Categorical	Monday - Sunday	Day factor and demand variance
previous_waste_kg	Numerical	0.0 - 120.0 kg	Historical waste from prior meal cycle
waste_percentage (Target)	Derived Target	2.0% - 48.0%	Operational waste ratio calibrated on training set

3. Preprocessing & Feature Engineering Pipeline

To prevent target leakage, all post-meal continuous measurements (*actual_waste_kg* and *waste_percentage*) were strictly excluded from feature inputs. The scikit-learn pipeline comprises:

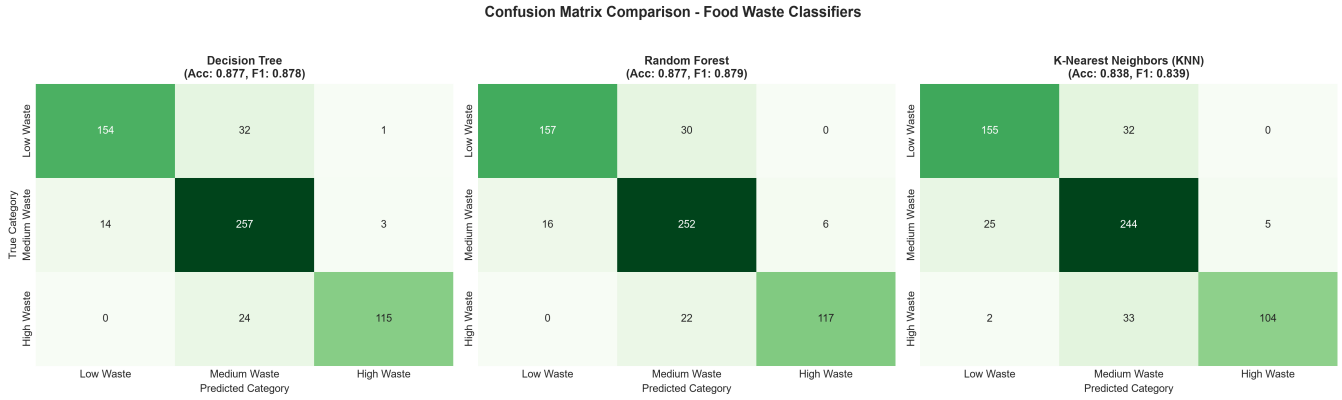
- **Domain Feature Engineering:** Computes prepared_per_person, prev_waste_ratio, and portion_excess_ratio relative to standard appetite baselines.
- **ColumnTransformer:** Applies OneHotEncoder(handle_unknown='ignore') for categoricals and StandardScaler() for numerical features.
- **Stratified 80/20 Train-Test Split:** 2,400 training samples and 600 test samples partitioned with stratify=y and random_state=42.

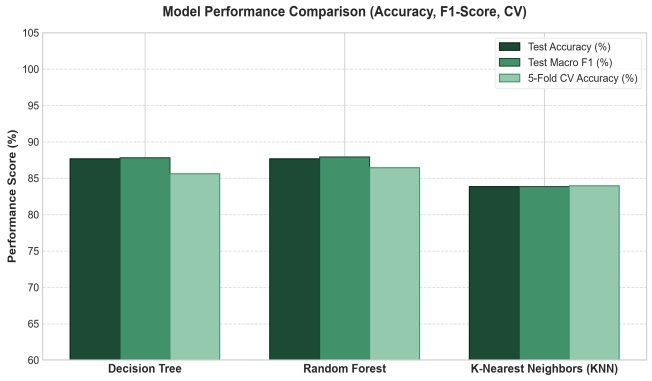
4. Machine Learning Algorithms & Benchmark Comparison

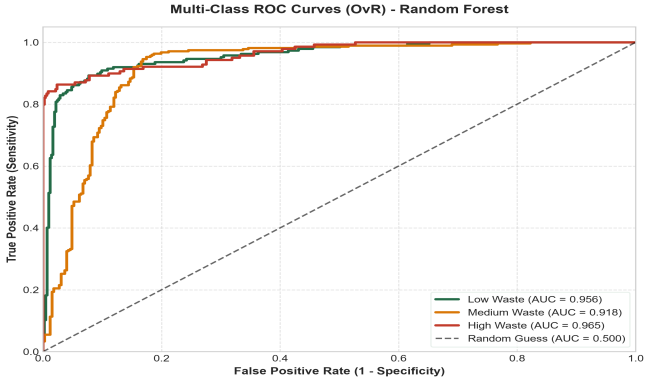
Three classification algorithms were compared under 5-fold stratified cross-validation on the training set. Linear Regression and Logistic Regression were excluded per project constraints.

Model Algorithm	5-Fold CV Acc.	Test Acc.	Macro Prec.	Macro Recall	Macro F1	Weighted F1	ROC-AUC
Random Forest (Selected)	**86.46%**	**87.67%**	0.8959	0.8670	**0.8791**	0.8770	**0.9464**
Decision Tree	85.62%	87.67%	0.9014	0.8629	0.8782	0.8768	0.9415
K-Nearest Neighbors (KNN)	83.96%	83.83%	0.8651	0.8225	0.8386	0.8384	0.9341

5. Experimental Evaluation Visualizations







6. Scenario Testing & API Verification

The model and FastAPI service (*app.py*) were tested across realistic operational scenarios and edge cases. Results confirmed consistent, logically sound category predictions:

Scenario Name	Meal Parameters	Prediction Output	Confidence	Status Badge
Optimal Lean Lunch	Lunch, 32 kg, 48 diners, 1.5 kg prev waste	**Low Waste**	82.0%	High Efficiency ■
Standard Buffer Lunch	Lunch, 38 kg, 48 diners, 4.0 kg prev waste	**Medium Waste**	87.7%	Standard Margin
Severe Dinner Overprep	Dinner, 85 kg, 40 diners, 18.0 kg prev waste	**High Waste**	99.7%	Surplus Alert ■■
Optimal Breakfast	Breakfast, 15 kg, 38 diners, 0.8 kg prev waste	**Low Waste**	83.0%	High Efficiency ■
Invalid / Negative Input	Lunch, -10 kg prep or 0 diners	**Rejected (HTTP 422)**	N/A	Schema Error

7. Conclusion & Project Limitations

Conclusion: The Random Forest classification pipeline demonstrated high predictive accuracy (87.67% Test Accuracy, 0.8791 Macro F1, 0.9464 ROC-AUC) across diverse kitchen operational settings, successfully bridging meal planning data with actionable waste mitigation.

Limitations: The dataset is a benchmark simulation derived from aggregate hospitality standards. In real-world cafeteria deployments, logging 30-60 days of site-specific smart bin telemetry is recommended to account for localized menu preferences.